

SCREENSHOT ORKESTRASI KUBERNETES

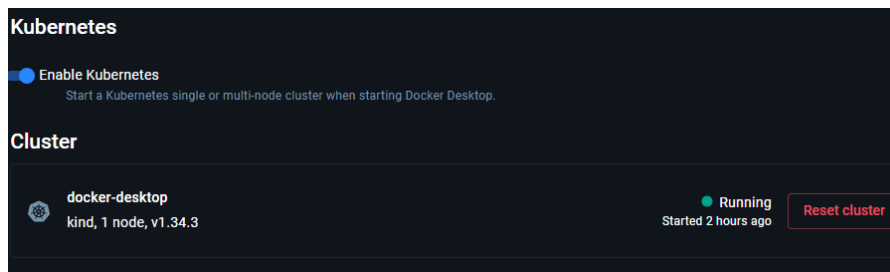
SOFTWARE ENGINEER-2

Nama : Andito Dwi Wicaksono

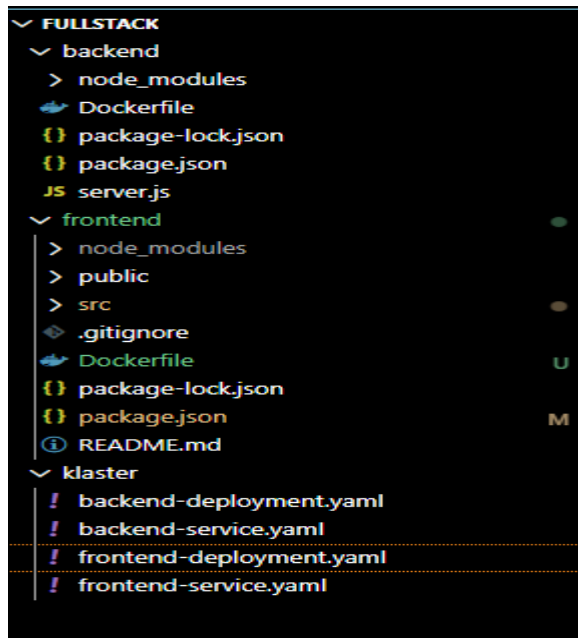
NIM : 2410511115

➤ Langkah Download Kubernet

```
C:\SEMESTER 4\SE-2\fullstack>kubectl version --client
Client Version: v1.34.1
Kustomize Version: v5.7.1
```



➤ Direktori Folder



➤ Langkah backend

Buat folder backend (mkdir biasa) lalu konfigurasi seperti biasa (npm init -y) dan (npm install express) dan Membuat Server menggunakan Express.js dengan menggunakan nama file server.js

a. echo > Dockerfile

```
C:\SEMESTER 4\SE-2\fullstack\backend>Type Dockerfile
FROM node:18
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .
EXPOSE 3011
CMD ["node", "server.js"]
```

➤ Langkah Frontend

Dalam membuat folder frontend tidak bisa menggunakan mkdir biasa, yang perlu dilakukan adalah menggunakan, command untuk membuat project react “npm create-react-app frontend”

```
C:\SEMESTER 4\SE-2\fullstack>npx create-react-app frontend
Need to install the following packages:
create-react-app@5.1.0
Ok to proceed? (y) y
```

a. Echo >Dockerfile

```
C:\SEMESTER 4\SE-2\fullstack\frontend>Type Dockerfile
FROM node:18 as build
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .
RUN npm run build

FROM nginx:alpine
COPY --from=build /app/build /usr/share/nginx/html
EXPOSE 80
```

➤ Build & Push

a. Backend

```
C:\SEMESTER 4\SE-2\fullstack>docker build -t anditodwiii/backend:v1 ./ba
ckend
[+] Building 97.5s (11/11) FINISHED          docker:desktop-linux
=> [internal] load build definition from Dockerfile      0.1s
=> == transferring dockerfile: 171B                      0.0s
=> [internal] load metadata for docker.io/library/node:18 5.6s
=> [auth] library/node:pull token for registry-1.docker.io 0.0s
```

```
C:\SEMESTER 4\SE-2\fullstack>docker push anditodwiii/backend:v1
The push refers to repository [docker.io/anditodwiii/backend]
5ec4f5daef83: Waiting
37927ed901b1: Waiting
78172c3b56a3: Waiting
33907d18a1c0: Waiting
96fd7b9f5fc7: Waiting
cda7f44f2bdd: Waiting
3697be50c98b: Waiting
```

b. Frontend

```
C:\SEMESTER 4\SE-2\fullstack>docker build -t anditodwiii/frontend:v1 ./frontend
[+] Building 18.6s (7/15) docker:desktop-linux
=> [internal] load build definition from Dockerfile 0.1s
=> => transferring dockerfile: 239B 0.0s
=> WARN: FromAsCasing: 'as' and 'FROM' keywords' casing do not m 0.1s
```

```
C:\SEMESTER 4\SE-2\fullstack>docker push anditodwiii/frontend:v1
The push refers to repository [docker.io/anditodwiii/frontend]
94d083cf706a: Waiting
```

c. Hasil push dockerhub

☐	☐	anditodwiii/backend	v1	965f00df81f1	27 minutes ag	1.57 GB
☐	☐	anditodwiii/frontend	v1	947db2fd54ea	14 minutes ag	94.24 MB

➤ Langkah klaster

- eksekusi atau buat Klaster menggunakan comman Kuberentesnya (kubectl apply -f folder/file)

```
C:\SEMESTER 4\SE-2\fullstack>kubectl apply -f klaster/backend-deployment.yaml
deployment.apps/backend created
```

```
C:\SEMESTER 4\SE-2\fullstack>kubectl apply -f klaster/backend-service.yaml
service/backend-service created
```

```
C:\SEMESTER 4\SE-2\fullstack>kubectl apply -f klaster/frontend-deployment.yaml
deployment.apps/frontend created
```

```
C:\SEMESTER 4\SE-2\fullstack>kubectl apply -f klaster/frontend-service.yaml
service/frontend-service created
```

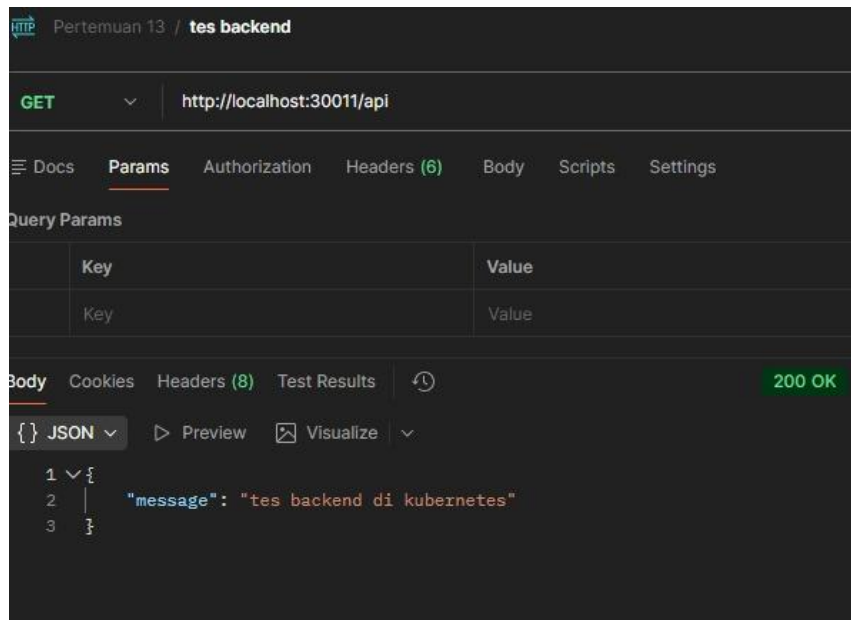
- b. cek port dan service yang terbentuk dengan command `kubectl get svc`

```
C:\SEMESTER 4\SE-2\fullstack>kubectl get svc
```

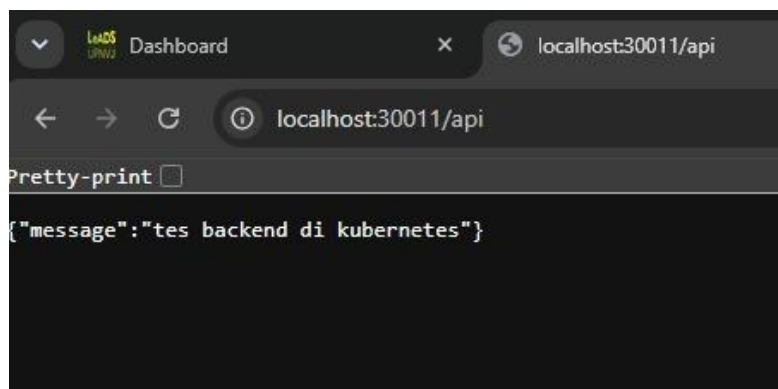
NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
backend-service	NodePort	10.96.201.21	<none>	3011:30011/TCP	53s
frontend-service	NodePort	10.96.40.33	<none>	80:30080/TCP	4m3s
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	111m

- Langkah Menguji Backend dan Frontend yang sudah terklaster dan terhubung

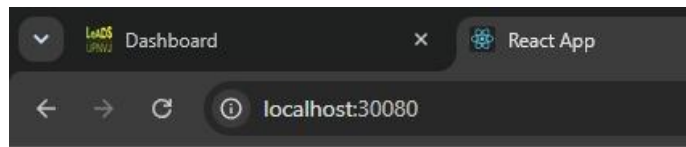
- a. Tes Backend di Postman



- b. Tes backend di Browser



- c. Uji Frontend yang sudah terhubung dengan Backend dengan klaster yang terbentuk dengan link : `localhost:30080`



Tes Frontend

coba tes dari backend: tes backend di kubernetes

- d. mengecek service yang aktif

```
C:\SEMESTER 4\SE-2\fullstack>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
backend-66bc89789c-hx9qv            1/1     Running   0           20m
frontend-7d5684b446-kfn56          1/1     Running   0           19m
```